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Abstract

1 Introduction

2 Preliminaries

3 How OthelloGPT Encodes Rules

In this section, we introduce the rules of Othello, rigorously define the 3 classes of illegal moves, and show that OthelloGPT follows these rules.

The game rules of Othello are summarized as follows:

- **R1:** The player must place a disc on a currently empty square.
- **R2:** The player's disk must flank a contiguous line of their opponents pieces (horizontally, vertically, and/or diagonally). After the move, the opponent's disks turn in the player's color.
- **R3:** The player's move is only valid if it is the player's turn. Players take turns moving unless a player cannot play a move that flanks their opponent's discs (per R2), in which case their turn is skipped.

The above rules of Othello simultaneously define 3 illegal move classes. The illegal class **R1** encloses illegal moves t that *place disks on squares already occupied.* Similarly, illegal class **R2** encloses illegal moves that are placed on empty squares but *do not flank the opponent's discs.* Finally, **R3** encloses illegal moves that *would have been legal* if it were actually the opponent's turn.

See Figure 1 (a) for rigorous examples of R1, R2, and R3 in play. In (b), we find that the probability any move played was illegal up to t grows exponentially, but always remains low (under 10%).

Additionally, we train a tensor product probe coupling the 64 squares with the relative disc colors $\{\text{empty}, \text{theirs}, \text{mine}\}$. See Figure 2 – we find that OthelloGPT accurately updates the relative disc colors after the flanking is was played.

4 Probes Reveal Geometry of Legal & Illegal Moves in Model Activations

In this section, we first show that the rules of Othello can be accurately decoded from the model's residual stream activations using two linear probes: the first is trained to classify moves as $\{\text{legal}, \text{illegal}\}$, and the second $\{\text{legal}, \text{R1}, \text{R2}, \text{R3}\}$ – see Figure 3 (c).

Second, we show that the five aggregate linear probe directions are geometrically separable via 2D Multidimensional Scaling (MDS). MDS takes in the matrix of geodesic angles $\theta_{ij} = \arccos(x_i^\top x_j)$ between unit-normalized probe directions, and places these geodesic angles in 2D space such that

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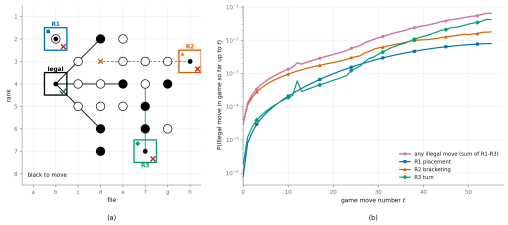


Figure 1: **OthelloGPT follows the rules of Othello (a)** Randomly chosen game, board state at move $t = 16$, black to move. Examples of a legal move and the 3 types illegal moves (R1, R2, R3) are colored on the board. **(b)** Probability that any move was illegal up to the current move number t, differentiated by the illegal move classes (averaged across $n = 1000$ OthelloGPT rollout games).

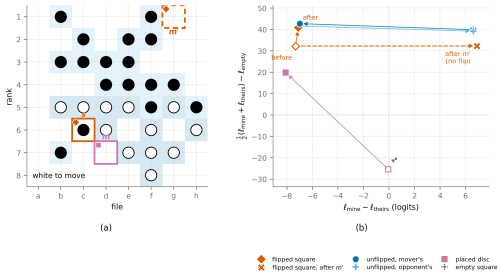


Figure 2: **OthelloGPT understands that flanking flips opponent's discs (a)** Randomly chosen game, board state at move $t = 25$, white to move. 'm' denotes move $t = 26$, 's' denotes opponent's flanked disc, and 'm" denotes the matched control move. **(b)** We train a tensor product probe (TPR) on the board state of Othello. The x-axis, $\ell_{\text{mine}} - \ell_{\text{theirs}}$, is a subtraction of the 'mine' and 'theirs' directions of the TPR, and shows what OthelloGPT thinks are the relative disc colors. The y-axis, $\frac{1}{2}(\ell_{\text{mine}} + \ell_{\text{theirs}}) - \ell_{\text{empty}}$, shows whether OthelloGPT thinks the squares are empty or occupied.

their Euclidean distances correlate with how close the directions are. As such, we find as evident in Figure 3 (a) that OthelloGPT learns unique directions for each of the five classes and geometrically separates them from each other.

Third, cosine similarity between the probe directions demonstrates that OthelloGPT understands how similar or dissimilar each of the five directions are to each other. For example, in Figure 3 (b), $\{\text{legal}, \text{illegal}\}$ score -0.83, meaning that OthelloGPT encodes legal and illegal in nearly opposite directions. Additionally, we find that $\{R1, R2, R3\}$ all score < 0 with each other. This makes sense because, for instance, if an illegal move falls under violation $R1$, it cannot simultaneously fall under $R2$ or $R3$.

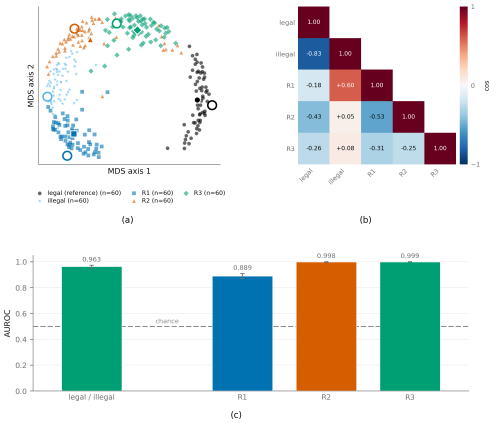


Figure 3: **Rule violations can be accurately decoded and geometrically separated using linear probes. (a)** 2D Multidimensional Scaling (MDS) of linear probe directions (ring is pooled across squares, point is per-square) across five classes: legal, illegal, R1, R2, and R3. **(b)** Cosine similarity matrix between the five probe directions. **(c)** Probes' AUROC across the five directions.

5 Illegal Move Detector

In this section, we propose a novel metric that detects whether a move in a game violated a rule.

Let $z_t \in \mathbb{R}^{512}$ denote OthelloGPT's layer-5 residual stream activations at the position of move t, and let

$$d_t = \|z_t - z_{t-1}\|_2 \tag{1}$$

be the norm step length at move t. We standardize the norm step length for each $t \in [2, 58]$ against a corpus of $n = 2000$ legal games:

$$g_t = -\frac{d_t - \mu_t}{\sigma_t}, \qquad \mu_t = \mathbb{E}_{\text{legal}}[d_t], \quad \sigma_t^2 = \text{Var}_{\text{legal}}[d_t]. \tag{2}$$

Finally, for $t \in [6, 55]$, we find the normalized difference in g_t between the moves after t inclusive, i.e. $[t, t+m]$, against the moves before t exclusive, i.e. $[t-m-1, t-1]$ (moving sum):

$$S_t = \sqrt{\frac{m}{2}} \left(\frac{1}{m} \sum_{i=0}^{m-1} g_{t+i} - \frac{1}{m} \sum_{i=1}^{m} g_{t-i} \right), \qquad m = 4. \tag{3}$$

We do a moving sum because we are asking "was there a rule violation at t?", and we answer this question by measuring the causal affect that t has on g_t at a window of $m = 4$ before and after t.

In Figure 4, we show our metric, S_t, in action on a specific example game. In that game, we find that S_t peaks at t_0, the illegal move number, and correctly classifies $4/5$ of the window of moves as illegal or legal – see (a) and (b).

We also measure the AUROC of using our metric S_t at detecting $\{\text{illegal}, \text{R1}, \text{R2}, \text{R3}\}$ against $\{\text{legal}\}$. See Figure 4 (c) – we find that our metric achieves an AUROC of ~ 0.68 for detecting illegal moves, which is considerably better than the random baseline of 0.50. Within specific rule violations, our metric performs especially well at detecting R2 violations, achieving an AUROC of ~ 0.76, and performs less well at detecting R1 violations, achieving an AUROC of ~ 0.60 (though still above the random baseline). This difference in AUROC implies that our metric, at least for the model's residual stream activations, ascribes greater weight to the rule that moves must flank the opponent's discs than to the rule that discs must be played on empty squares.

Note: AUROC is the probability that a randomly picked violation scores higher than a randomly picked legal move. We therefore do not need to choose threshold(s) for S_t, making our metric completely ***unsupervised.***

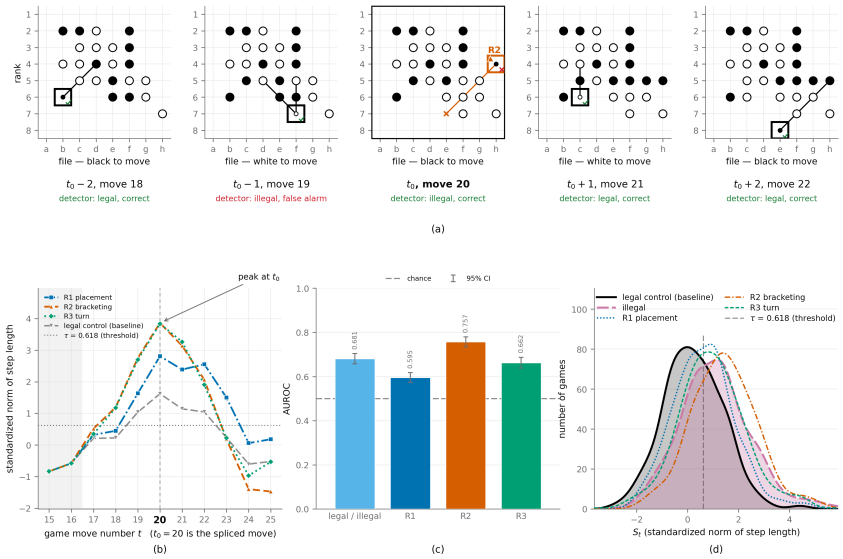


Figure 4: **Detecting rule violations with residual stream activations. (a)** Sequence of 5 moves in an Othello game centered at $t_0 = 20$, the illegal move (R2). **(b)** Value of detection metric (standardized norm of step length) in *residual stream activations* over a window of moves in the same game. **(c)** AUROC of using our metric to detect $\{\text{illegal}, \text{R1}, \text{R2}, \text{R3}\}$ against $\{\text{legal}\}$, averaged among $n = 500$ games. **(d)** Distributions of the detection metric across each of the five classes, averaged across $n = 500$ games.

5.1 SAEs Improve the Detector

While our metric performs reasonably well on the model's residual stream activations, in this section, we test our metric on SAEs, and find that half of the SAEs we tested – namely GatedSAE, BatchTopK, TopK, and GatedAnneal – achiever higher AUROC than that of the residual stream. GatedSAE achieves the highest AUROC, – ~ 0.75 – for detecting illegal from legal moves (see Figure 5 (a)).

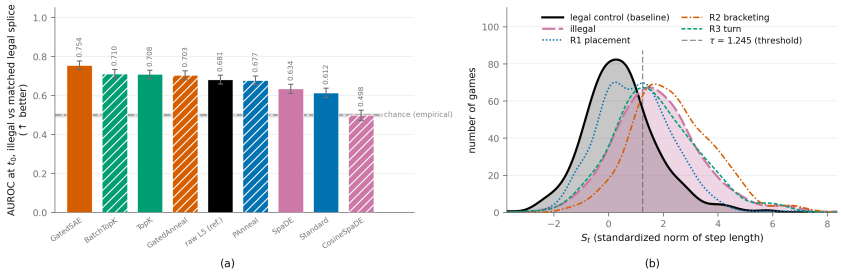


Figure 5: **Detecting rule violations with SAEs. (a)** AUROC of using our metric in *SAE latent activations* to detect $\{\text{illegal}, \text{R1}, \text{R2}, \text{R3}\}$ against $\{\text{legal}\}$, averaged among $n = 500$ games, averaged among $n = 500$ games. **(b)** Distributions of the detection metric in the SAE latent activations of the SAE that scored the highest AUROC – GatedSAE – across each of the five classes, averaged across $n = 500$ games.

References

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

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A Experimental Setup

B Further Results